

Self-Adjointness of the Fundamental Fractal Operator $\hat{F}$

Closure of the R82 gap from Doc. 322

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Theorem 3: " $\varphi_k^{\text{COV}} = \sum r_n \cos \theta_{n,k} \in \mathbb{R}$, hence $\hat{F} = \hat{F}^\dagger$ " → under (F2), $\hat{F}$ is normal; $\hat{F}$ is self-adjoint on the null sectors P_0 , on charged pairs $\hat{F}^\dagger$ is the $\mathbb{Z}_3$ -mirrored $\hat{F}$, and the self-adjoint operator there is $\text{Re } \hat{F} = (\hat{F} + \hat{F}^\dagger)/2$ (postulate **[S]**); the counter-example is dropped.
- Main result, Corollaries 4 and 7, status table: "fully closed, with no remaining cases" → symmetry **[B]** on P_0 , **[S]** on the whole space (cf. Doc. 190, R145 "R86 refined"); verification script 21 → 24 assertions.

Status markers: **[B]** algebraically proved; **[K]** derivable from ξ or numerically checked; **[S]** sketched or postulated, not proved (open).

Abstract

Doc. 322 defines the fundamental fractal operator

$$\hat{F} = \bigoplus_{n=1}^N S_n \circ L_n, \quad S_n = r_n U_n, \quad r_n \in (0, 1), \quad U_n \text{ unitary}, \quad (.1)$$

with logical projectors $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$, and books self-adjointness as an open sketch **[S]** (R82).

The upper bound N in (.1) is *finite*: $L_0 = \xi \cdot \ell_P$ (Doc. 180 **[K]**) is the minimal length scale of FFGFT — below L_0 the concept of "distance" loses physical meaning. The $\bigoplus_{n=1}^\infty$ notation in Doc. 322 is a functional-analytic convention, not a physical infinity.

Main result [B] on P_0 , [S] on all of $\mathcal{H}$

The R82 gap is **closed on the $\mathbb{Z}_3$ -invariant sector P_0** ; on the whole space it remains open **[S]**. Because L_0 makes the number of rungs finite, $\hat{F}$ is bounded:

$$\|\hat{F}\| \leq \sum_{n=1}^N r_n < \infty.$$

A bounded symmetric operator is automatically self-adjoint; the deficiency indices are $(0,0)$; there is exactly one self-adjoint realisation. The $\mathbb{Z}_3$ pairing $(k, -k)$ makes $\hat{F}$ normal; on P_0 , $\hat{F} = \hat{F}^\dagger$. On charged pairs $\hat{F}^\dagger$ is the $\mathbb{Z}_3$ -mirrored $\hat{F}$; the self-adjoint operator there is $\text{Re } \hat{F} = (\hat{F} + \hat{F}^\dagger)/2$ — a postulate **[S]**. $\xi = \lambda_{\min}(\hat{F}_{D_4})$ is well-defined provided the D_4 sector lies in the $\mathbb{Z}_3$ -invariant sector **[B]**. (Corrected on 1 Oct. 2026: previously “fully closed, with no remaining cases” and “The symmetry $\hat{F} = \hat{F}^\dagger$ follows from the $\mathbb{Z}_3$ pairing, not an extra requirement” — with $\varphi_{-k} = \bar{\varphi}_k$, $\hat{F}$ is only normal, $\|\hat{F} - \hat{F}^\dagger\| = 2|\text{Im } \varphi_k| \neq 0$ on charged pairs; cf. Theorem 3 and Doc. 190, R145 “R86 refined”.)

Verification script: 327_selbstadjungiertheit_F_verify.py — 24 assertions, all passed.

.1 Starting point

R82 (Doc. 190) records: “Open: complete self-adjointness proof for $\hat{F}$ **[S]**.” Without this proof, the central claim

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}$$

is formally unsecured: “smallest eigenvalue” requires a real spectrum, and a real spectrum requires self-adjointness.

The definition in Doc. 322 contains a genuine tension: $S_n L_n = r_n U_n L_n$ is in general *not* symmetric for arbitrary unitary U_n — $(U_n L_n)^\dagger = L_n U_n^\dagger \neq U_n L_n$. The self-adjointness must come from the *structure*. This is exactly what the $\mathbb{Z}_3$ pairing delivers.

.2 Three observations from the corpus

The following points introduce no new axioms — they read from already-established corpus results what is needed for the proof.

(F1) Finite number of rungs. $L_0 = \xi \cdot \ell_P$ is the minimal length scale (Doc. 180 **[K]**). The scale ladder ends at L_0 ; N is finite; $\sum_{n=1}^N r_n < \infty$ follows trivially. No convergence condition needs to be imposed — there are no infinities in FFGFT.

(F2) $\mathbb{Z}_3$ phase pairing. U_n acts sector-wise as a phase rotation $e^{i\theta_{n,k}}$ (Doc. 322: “phase rotations on the scale sectors”), and the phases respect the orbifold symmetry: $\theta_{n,-k} = -\theta_{n,k}$ for $\mathbb{Z}_3$ -charged pairs $(k, -k)$; on null sectors ($q = 0$): $\theta \in \{0, \pi\}$. The same pairing carries the Hawking sector information in Doc. 325.

(F3) Finite fractal recursion. The fractal measure arises from a finite 100-step recursion (Doc. 322), hence $0 < c \leq w \leq C$ for the weight w — bounded and strictly positive.

3 Proof chain

Lemma 1 — Filtration realises the Doc.-322 axioms [B]

Under (F1), the L_n satisfy exactly $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$: the “fractal inclusion structure” of Doc. 322 is a scale filtration. (*Script [1]: exact to machine precision.*)

Theorem 2 — Diagonal form and boundedness [B]

Under (F1)+(F2), $\hat{F}$ is block-diagonal:

$$\hat{F}|_{\mathcal{H}_k} = \varphi_k \cdot \mathbf{1}, \quad \varphi_k = \sum_{n \geq k}^N r_n e^{i\theta_{n,k}}.$$

By the finite rung count (F1): $\|\hat{F}\| \leq \sum_{n=1}^N r_n < \infty$. $\hat{F}$ is **bounded**, $\mathcal{D}(\hat{F}) = \mathcal{H}$. (*Script [2]+[3].*)

Theorem 3 — $\mathbb{Z}_3$ pairing: normality [B], symmetry on P_0 [B]

Under (F2), $\hat{F}$ is normal (block-diagonal with scalars φ_k); $\varphi_{-k} = \bar{\varphi}_k$ holds when the sectors k and $-k$ carry the same scale rungs n . On the null sectors ($\theta \in \{0, \pi\}$), $\hat{F}$ is self-adjoint. On charged pairs $\hat{F}^\dagger$ is the $\mathbb{Z}_3$ -mirrored $\hat{F}$, $\|\hat{F} - \hat{F}^\dagger\| = 2|\operatorname{Im} \varphi_k|$; the self-adjoint operator there is

$$\operatorname{Re} \hat{F} = \frac{1}{2}(\hat{F} + \hat{F}^\dagger), \quad \varphi_k^{\operatorname{cov}} = \sum_{n \geq k}^N r_n \cos \theta_{n,k} \in \mathbb{R}$$

(cosine form) — a postulate **[S]**. (*Script [4]: normality, $\|\hat{F} - \hat{F}^\dagger\| = 0.019$ on charged pairs, hermiticity on P_0 .) (Corrected on 1 Oct. 2026: previously “The orbifold-covariant operator has real sector values $\varphi_k^{\operatorname{cov}} = \sum r_n \cos \theta_{n,k}$, hence $\hat{F} = \hat{F}^\dagger$. Symmetry is a $\mathbb{Z}_3$ consequence, not a requirement. Counter-example: one unpaired phase gives $\|\hat{F} - \hat{F}^\dagger\| \approx 0.02$ ” – by Theorem 2, φ_k is complex; the pairing only gives $\varphi_{-k} = \bar{\varphi}_k$, i.e. normality; the cosine form is $(\hat{F} + \hat{F}^\dagger)/2$, a different operator. In the script model $\max |\hat{F} - \hat{F}^\dagger| = 0.0193$ with pairing, the same as in the former counter-example without pairing, so the counter-example does not discriminate.)*

Corollary 4 — Self-adjointness, deficiency indices $(0, 0)$ [B] on P_0

Bounded (Thm. 2) + symmetric (Thm. 3) $\Rightarrow$ self-adjoint: for $\hat{F}$ on $P_0 \mathcal{H}$, for $\operatorname{Re} \hat{F}$ on all of $\mathcal{H}$ (there a postulate **[S]**). Deficiency indices: $\ker(A \mp i) = \{0\}$ for $A = \hat{F}|_{P_0 \mathcal{H}}$ or $A = \operatorname{Re} \hat{F}$, since $\|(A \mp i)^{-1}\| \leq 1$ for bounded self-adjoint A . **Exactly one** self-adjoint

realisation — no extension choice. (*Script [5]: smallest singular value exactly 1, for $\text{Re } \hat{F}$.*) (Corrected on 1 Oct. 2026: previously “self-adjoint on $\mathcal{H}$ ” for $\hat{F}$ itself – Theorem 3 gives symmetry of $\hat{F}$ only on P_0 .)

Theorem 5 — Tensor factor $\hat{\Delta}_{T^4}$ with fractal measure [B]

1. Δ on compact T^4 (no boundary): essentially self-adjoint, Fourier eigenbasis with eigenvalues $|n|^2 \geq 0$. (Standard.)
2. Fractal measure (F3): $0 < c \leq w \leq C$, so $\psi \mapsto \sqrt{w}\psi$ is a unitary isomorphism $L^2(d\mu_f) \rightarrow L^2(d\mu)$. The fractal measure preserves self-adjointness. (*Script [6].*)

Theorem 6 — $\mathbb{Z}_3$ restriction and χ -twists [B]

1. $[\hat{F}, P_0] = 0$ ($\mathbb{Z}_3$ covariance from (F2)) $\Rightarrow$ restriction to the orbifold sector is self-adjoint. (*Script [7].*)
2. Fixed-point boundary conditions $\chi^3 = 1$ (Docs. 322/320): unitarily implemented twists; spectrum $(n + \alpha)^2 \geq 0$ real in all three twist classes — each carries a self-adjoint realisation. (*Script [8].*)

Corollary 7 — ξ as well-defined minimum [B]

$\hat{F}_{D_4}$ (restriction to the D_4 sector, Doc. 314) is self-adjoint, provided the D_4 sector lies in the $\mathbb{Z}_3$ -invariant sector $P_0\mathcal{H}$ (otherwise the statement holds for $\text{Re } \hat{F}$, postulate [S]). By the min-max principle:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \min_{\|\psi\|=1, \psi \in \mathcal{H}_{D_4}} \langle \psi, \hat{F}\psi \rangle$$

well-defined and truncation-stable: the lowest mode lies in every truncation space that contains it. (*Script [9]: ξ is inserted in the model; what is checked is existence and truncation stability of the minimum.*) (Corrected on 1 Oct. 2026: previously without the condition “ D_4 sector in the $\mathbb{Z}_3$ -invariant sector” – self-adjointness of $\hat{F}$ is proved only there, cf. Theorem 3.)

.4 Status table

Statement	Status
$L_n L_m = L_{\min}$ realised by filtration, exactly	[B]
$\hat{F}$ bounded by finite rung count (L_0)	[B]
$\hat{F}$ normal; symmetry on P_0 as $\mathbb{Z}_3$ consequence	[B]
Symmetry on the whole space ($\text{Re } \hat{F}$ as postulate)	[S]
Self-adjointness, deficiency indices $(0, 0)$, uniqueness on P_0	[B]
Fractal measure (finite recursion) preserves self-adjointness	[B]
$\mathbb{Z}_3$ restriction and all three χ -twist classes s.a.	[B]
$\xi = \lambda_{\min}(\hat{F}_{D_4})$ well-defined, truncation-stable (D_4 in P_0)	[B]
Remaining cases	whole space open [S]

(Corrected on 1 Oct. 2026: previously “Symmetry as $\mathbb{Z}_3$ consequence, with counter-example **[B]**” and “Remaining cases: none” – symmetry is proved only on P_0 , cf. Theorem 3.)

The R82 gap is closed on the $\mathbb{Z}_3$ -invariant sector; on the whole space it remains open **[S]** (cf. Doc. 190, R145 “R86 refined”). (Corrected on 1 Oct. 2026: previously “The R82 gap is fully closed” – cf. Theorem 3.) There are no infinities in FFGFT; $\sum r_n < \infty$ was never an extra requirement — it follows directly from $L_0 = \xi \cdot \ell_P$.

Relation to the corpus

Building block	Doc.
Minimal length scale $L_0 = \xi \cdot \ell_P$	180
Definition of $\hat{F}$, $\mathcal{D}(\hat{F})$, R82 gap	322
D_4 lattice in Hilbert space	314
Fixed-point boundary conditions, neutrino localisation	320, 322
$\mathbb{Z}_3$ triality, sector structure	321
Sector pairing $(k, -k)$ — same as in Hawking mechanism	325
Fractal measure, 100-step recursion, $D_f = 3 - \xi$	322

References

1. J. Pascher, Docs. 180, 314, 320, 321, 322, 325, FFGFT corpus, 2026.
2. M. Reed, B. Simon, *Methods of Modern Mathematical Physics I/II*, Academic Press.
3. J. von Neumann, *Allgemeine Eigenwerttheorie Hermitescher Funktionaloperatoren*, Math. Ann. 102, 1929.

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